

Supplementary Information

An open-source, 3D-printed auger powder doser designed with generative artificial intelligence

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S1 Bill of materials

Table S1 lists the parts for one channel, plus the items a platform needs only once whatever its number of channels. Prices are single-unit US dollars as of September 2026 and should be checked at order time; vendor links are kept in `hardware/` in the repository (<https://github.com/vertical-cloud-lab/powder-doser>).

Table S1: Bill of materials. Printed parts are excluded (about 260 g of PLA per channel, under \$10), as are the custom circuit board, which can be replaced by a breadboard, and the analytical balance (A&D HR-100A, 0.1 mg readability), which most laboratories already own.

Part	Qty	Price (USD)	Source
<i>Per channel</i>			
NEMA-11 bipolar stepper, 0.67 A, 10 N·cm (11HS18-0674S)	1	14	StepperOnline
Pololu Tic T500 stepper controller	1	33	Pololu #3135
Pololu 33 V shunt regulator (clamps stepper back-voltage)	1	15	Pololu #3776
12 V push–pull solenoid, JF-0530B type (tapping)	1	7.50	Adafruit #412
DRV8871 H-bridge breakout (solenoid driver)	1	7.50	Adafruit #3190
ERM coin vibration motor, 10 mm	1	1.95	Adafruit #1201
DRV2605L haptic driver breakout (I ² C)	1	7.95	Adafruit #2305
MG996R-class metal-gear servo (tilt)	2	24	generic
Fasteners (M3), bulk capacitors	—	5	generic
<i>Shared (one per platform)</i>			
Raspberry Pi Pico W	1	6	Raspberry Pi
Waveshare Pico-2CH-RS232 board (balance interface)	1	9	Waveshare 19979
Mean Well GST60A12-P1J 12 V/5 A supply, cord, and pigtail	1	26	Digi-Key
Pololu D24V22F5 5 V buck regulator	1	19	Pololu #2858
Headers and wiring	—	10	generic
Total, single-channel platform		≈185	

S2 Construction guide

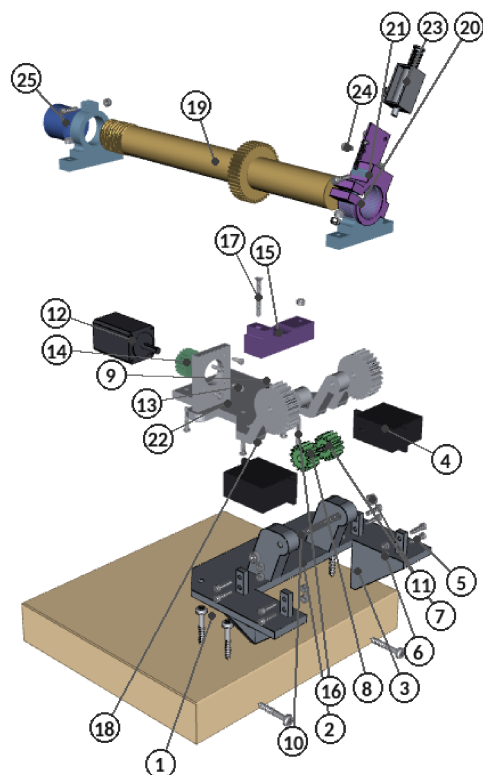
Table S2 lists the printed parts of the current doser and where to download each one. Fig. S1 shows every part of the mechanical assembly, numbered, with its bill of materials, and Fig. S2 shows the build in 19 steps. Both come from a CAD assembly of the team’s part files, which is also a public Onshape document (<https://cad.onshape.com/documents/ae9f107d3972fc9d390e541f>); the same steps are animated in the repository README. The wiring guide and firmware are in `hardware/` in the repository. The build, in outline (step numbers refer to Fig. S2, item numbers to Fig. S1):

1. **Print** the parts in Table S2 in PLA (0.2 mm layers, four perimeters, 25% infill). Most parts need support material. Print the auger upright.

2. **Base** (steps 1–5): screw the baseplate to a flat board or bench top (the CAD assumes 38 mm, the depth its front legs are made for), fit the two servos into its posts, and fit a 14-tooth pinion on each servo.
3. **Tilt stage** (steps 6–7): mount the mounting plate on the hinge with M5 screws and locknuts, so that the servo pinions mesh with the plate’s 28-tooth gears (2:1).
4. **Drive** (steps 8–10): bolt the NEMA-11 stepper to its plate, fit the 20-tooth pinion, and bolt the tap-collar base to the plate.
5. **Auger** (steps 11–17): slide one bracket and then the tap collar onto the tube from the outlet end, lower the auger onto the plate so that its 44-tooth gear meshes with the stepper pinion (2.2:1), slide the second bracket on from the cap end, and tighten the bracket screws and the tap-collar clamp. The collar’s ears rest against the raised stop on its base, so the collar stays put while the tube turns inside it.
6. **Solenoid** (step 18): screw the solenoid to the tap collar so that its plunger strikes the tube. (A vibration motor was also planned for the collar but was not used in this work.)
7. **Electronics**: wire the Pico W, Tic T500 (serial), DRV8871, DRV2605L (I²C), the RS-232 board (to the HR-100A balance), and the servos as shown in the schematic in `hardware/test-module/`, then flash the MicroPython firmware.
8. **Fill** (step 19): with the auger standing upright in the printed filling stand, unscrew the cap, pour the powder into the upper end of the tube, and screw the cap back on. There are no loading slots; the tube is its own reservoir. For small or costly samples, one of the two smaller augers in Table S2 can take its place.
9. **Checkout**: run the firmware self-test (rotate, tap, tilt, tare, and read), then the test protocols listed in the main text.

Table S2: Printed parts of the current doser and their design files. Every part except the tap-collar base was modelled by the team in Fusion 360, most of them as recreations of AI-modelled parts, and is shared publicly with downloads enabled. The tap-collar base is still the AI-modelled part, written by the coding agent in May 2026 (pull request #51). The auger, its cap, and the stepper pinion were on the rig for every test in the main text. STEP exports of the Fusion 360 parts are also in the repository (`cad/full-assembly/components/fusion-step/`, pull request #170), and the AI-modelled originals are in `cad/`.

Part	Design file
Auger: threaded storage tube with 44-tooth gear	https://a360.co/4y1oz2H
Screw-on cap	https://a360.co/4w9kRE5
Stepper pinion, 20 teeth	https://a360.co/4yqSHFz
Auger brackets (two)	https://a360.co/46XtYN1
Tap collar	https://a360.co/4AIIgyz
Tap-collar base (AI-modelled)	<code>mount_plate.step</code> , pull request #51
Mounting plate	https://a360.co/4xXUj8L
Baseplate	https://a360.co/4A0A7sI
Servo pinions, 14 teeth (two)	https://a360.co/4dcGSdF
<i>Accessories</i>	
Auger filling stand	https://a360.co/4yqMPfp
Smaller auger, about 3.3 mL	https://a360.co/4hxxCu5
Smaller auger, about 9.2 mL	https://a360.co/47qXATc



Bill of materials

Exploded view

No.	Part	Qty	Type
1	Mounting board	1	yours
2	Baseplate	1	printed
3	#10×1¼ in pan head wood screw	6	fastener
4	Servo	2	purchased
5	M3×14 socket head screw	10	fastener
6	M3 nylon-insert locknut	16	fastener
7	Servo pinion	2	printed
8	M3×10 socket head screw	2	fastener
9	Mounting plate	1	printed
10	M5×45 button head screw	2	fastener
11	M5 nylon-insert locknut	2	fastener
12	Stepper motor	1	purchased
13	M2.5×8 socket head screw	4	fastener
14	Stepper pinion	1	printed
15	Tap-collar base	1	printed
16	M3×25 button head screw	1	fastener
17	M3×30 flat head screw	1	fastener
18	M3 hex nut	1	fastener
19	Auger	1	printed
20	Bracket	2	printed
21	Tap collar	1	printed
22	M3×20 button head screw	5	fastener
23	Solenoid	1	purchased
24	M3×5 socket head screw	2	fastener
25	Auger cap	1	printed

Figure S1: Exploded view of the current design with every item of its mechanical bill of materials numbered: printed parts, purchased parts, and fasteners (McMaster-Carr part numbers in `cad/full-assembly/BOM.md`). Item 1 is whatever the doser is screwed to; the CAD uses a board 38 mm thick, the depth the baseplate's front legs are made for. The printed parts are the files in Table S2; the stepper, servos, and solenoid are simplified models made from their datasheets. Electronics are listed in Table S1. The CAD assembly was built by an AI agent from the team's part files, working through Onshape's programming interface, and rendered from the same files (pull request #170).

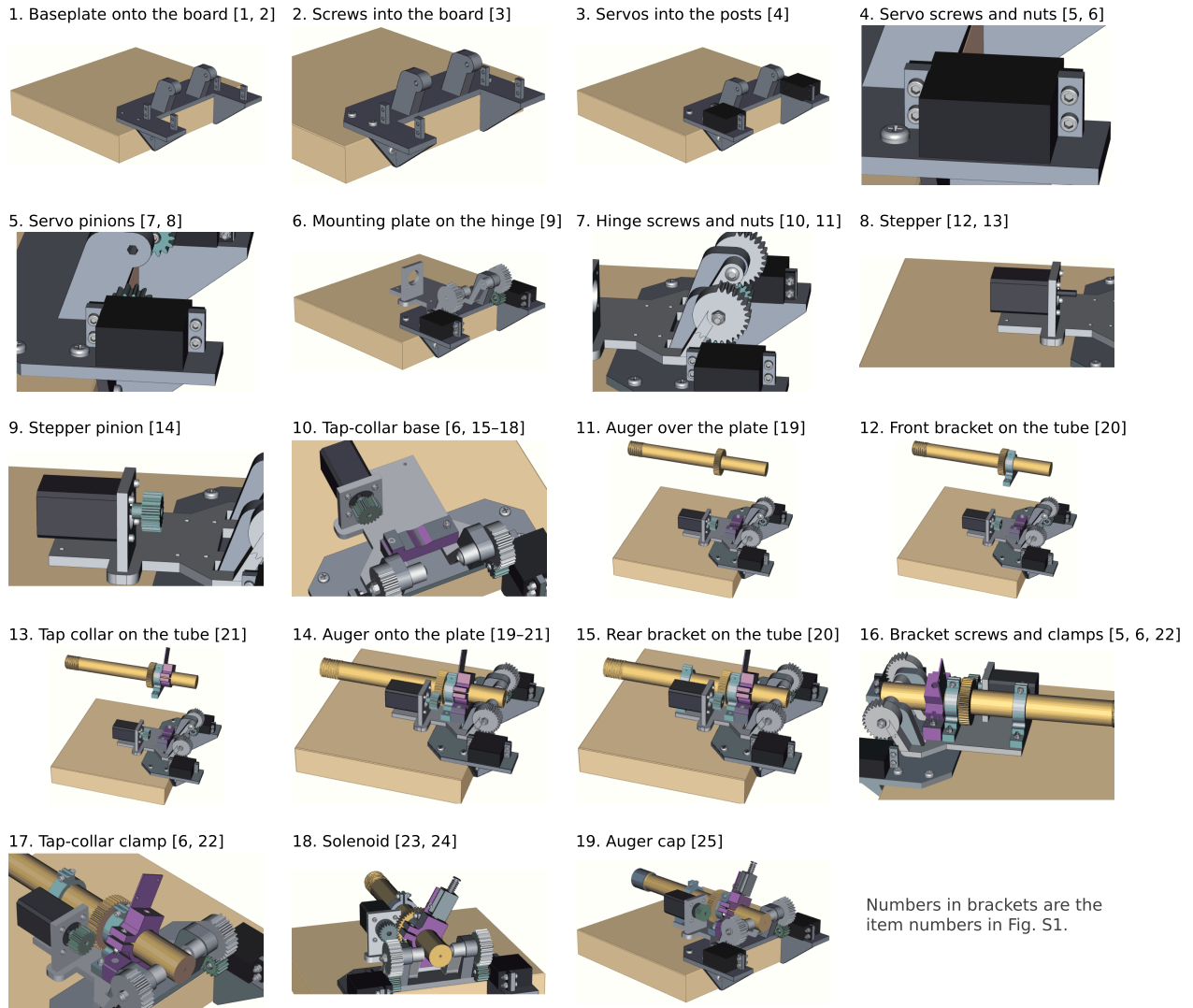


Figure S2: The build in 19 steps, as stills from the end of each step of the assembly animation in the repository. Bracketed numbers are the item numbers of Fig. S1. This order passes the CAD assembly's collision checks; it is one workable order rather than a record of how the test rig was built.

S3 Auger and outlet geometry

Fig. S3 shows the AI-modelled geared auger and its four printed outlet variants, made early in the project. The auger used in the tests is the team's later Fusion 360 part, shown in section in Fig. 1c of the main text; like the Type 4 variant, it ends in a tapered outlet with a 3 mm exit hole.

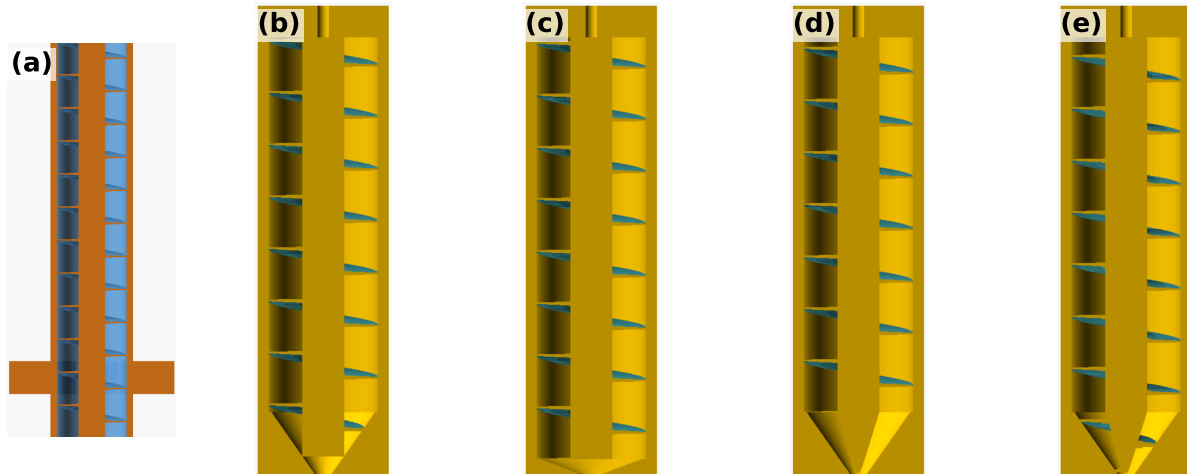


Figure S3: (a) Cross-section of the AI-modelled geared auger: the 8 mm central core, the single-start helical flight (10 mm pitch, 2 mm thick) running the full length of the 21 mm bore, and the external gear band. (b–e) Cross-sections of the four outlet variants (Types 1–4), which differ in exit constriction and chamfer. All are parametric OpenSCAD models (`cad/auger-geared/` in the repository) written by AI coding agents under human review; no conventional CAD software was used to design them. The auger used in the tests is the team's later Fusion 360 part (main text, Fig. 1c). Its flight is 0.5 mm thick with a 10.4 mm pitch and covers only the third of the tube nearest the outlet, the tube is filled through a screw-on cap rather than through loading slots, and its gear has 44 teeth rather than 48.

S4 Bench noise

Protocol A takes eight balance readings with nothing moving at the start of every run. Their spread sets the smallest change that run can resolve (Fig. S4). It was 0.0 mg on every run up to 12 August 2026 and rose to tens of milligrams from 20 August. Runs are compared against their own noise floor throughout the main text.

Noise after actuation

The controller acts on readings taken 1.5 s after it moves, so we also measured the noise of those readings (Fig. S5a). For each run, we took the robust standard deviation (1.4826 times the median absolute deviation) of the change between consecutive tap-phase readings. Tapping adds only a few milligrams per cycle and the cup cannot lose mass, so on the noisier runs this scatter is balance noise. On quiet runs it is an upper bound, because it also contains the powder that each tap delivered. The noise after actuation was at most 0.9 mg in all six first-round runs with a tap phase, but 5–19 mg in five of the ten second-round runs with small doses. On those same runs, the first reading of each dose, taken at rest, scattered by at most 0.4 mg, so the stability check made before each set of doses could not detect this noise. On the noisy runs, lowering the tube to horizontal alone shifted the reading by a median of 5–16 mg, which suggests that the doser's own motion disturbed the balance.

Table S3 compares each small dose's margin to its acceptance limit with the noise of its own run. Of the 14 doses of 50 mg that finished within the limit, only one lies more than twice its run's noise inside it, and 12 lie within one noise width of the limit. Of the 14 that missed, 12 lie more than twice the noise outside it. At 200 mg, 6 of the 18 passes and 10 of the 12 failures are robust in the same sense. The failures are therefore real, while most 50 mg passes cannot be told apart from noise. The controller also stops at the first reading inside its band, so on noisy runs the recorded masses lean high. All

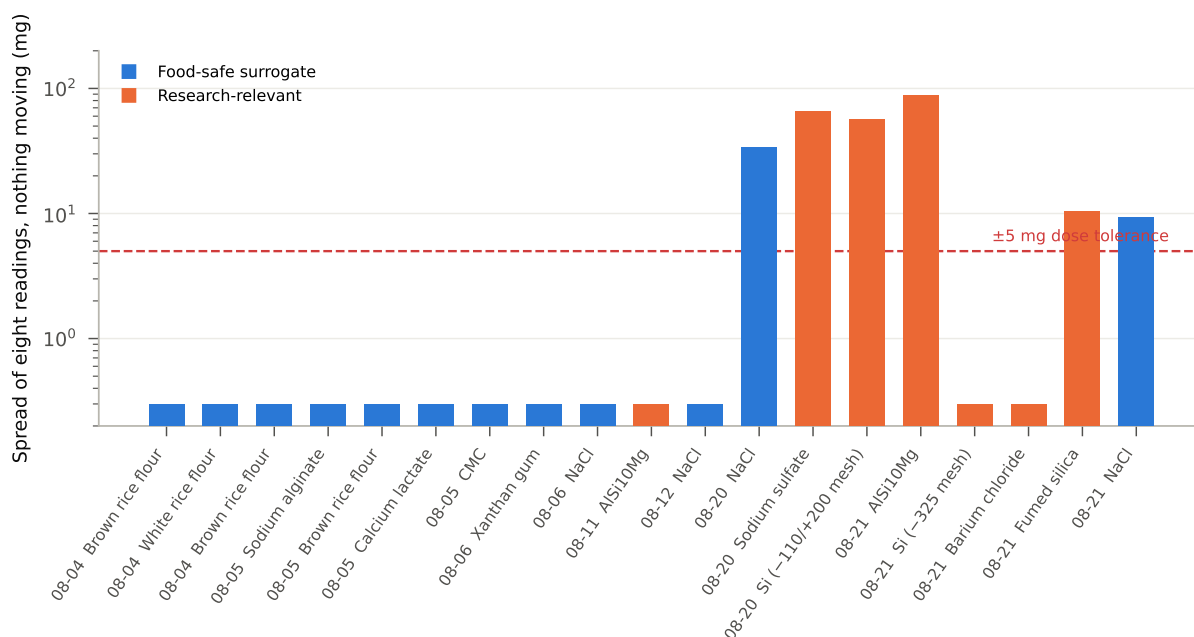


Figure S4: Balance noise floor for every first-round run: the spread (maximum minus minimum) of eight readings taken with nothing moving (protocol A), in date order. Runs whose eight readings were identical are drawn at the axis floor. The dashed line is the controller's ± 5 mg stopping band.

small doses come from the second round, in one fume hood with bracketed readings, so this comparison cannot be split by round or location.

Table S3: Pass/fail calls of the valid 50 and 200 mg doses compared with the balance noise after actuation in their own runs (Fig. S5). The margin is the acceptance limit minus the absolute dose error, and the noise is the robust standard deviation of the change between consecutive tap-phase readings. A call is robust when the margin exceeds twice the noise, marginal when it lies between one and two times the noise, and not resolvable otherwise. All these doses are from the second round, in the new fume hood, with bracketed readings. Per-dose values are in `paper/figures/data/post_actuation_noise_doses.csv`.

Target	Outcome	<i>n</i>	Robust	Marginal	Not resolvable
50 mg	within ± 5 mg	14	1	1	12
50 mg	outside ± 5 mg	14	12	0	2
200 mg	within ± 10 mg	18	6	3	9
200 mg	outside ± 10 mg	12	10	0	2

S5 Closed-loop dose record

Table S4 summarizes every valid closed-loop dose by powder and target mass. A dose is valid when its run passed quality screening, the dose ended in a defined controller state, and the auger or solenoid actually moved; doses excluded for any other reason (for example, a misaligned outlet, an unreadable balance, or powder that caked in the tube) are listed with the reason in the per-dose file.

Table S4: Closed-loop dose summary for every valid dose (protocol G at 1 g; protocol H at 50 and 200 mg). Error is delivered minus requested mass. *Within limit*: doses inside the acceptance limit ($\pm 10\%$ at 50 mg, $\pm 5\%$ at 200 mg and 1 g). *Firmware ok*: doses the controller ended inside its own ± 5 mg stopping band. Round 1: August 2026, balance read on its stability flag; round 2: September 2026, bracketed readings. Per-dose records, including invalid doses and the reason each was excluded, are in `paper/figures/data/doses_all.csv`.

Powder	Target	Round	<i>n</i>	Mean error (mg)	SD (mg)	Within limit	Firmware ok	Median time (s)
AlSi10Mg	50 mg	2	3	+21.5	26.7	1/3	1/3	15
Si (−110/+200 mesh)	50 mg	2	2	+1.1	4.8	2/2	2/2	18
Sodium sulfate	50 mg	2	3	+28.9	15.7	0/3	0/3	10
Calcium lactate	50 mg	2	2	+9.8	15.8	1/2	1/2	30
Xanthan gum	50 mg	2	3	+17.8	36.4	2/3	2/3	37
NaCl	50 mg	2	3	+4.2	4.8	2/3	2/3	17
CMC	50 mg	2	3	−2.2	2.4	3/3	3/3	211
White rice flour	50 mg	2	3	−4.7	0.6	3/3	3/3	136
Sodium alginate	50 mg	2	3	−19.6	8.3	0/3	0/3	561
Si (−325 mesh)	50 mg	2	3	−50.0	0.1	0/3	0/3	31
AlSi10Mg	200 mg	2	3	−0.3	2.7	3/3	3/3	214
Si (−110/+200 mesh)	200 mg	2	3	+9.8	19.8	2/3	2/3	34
Sodium sulfate	200 mg	2	3	+10.8	17.7	2/3	1/3	30
Calcium lactate	200 mg	2	3	−16.5	16.6	2/3	1/3	464
Xanthan gum	200 mg	2	3	+2.9	0.6	3/3	3/3	89
NaCl	200 mg	2	3	+2.8	9.2	2/3	2/3	53
CMC	200 mg	2	3	−22.6	23.4	1/3	1/3	903
White rice flour	200 mg	2	3	−4.9	0.1	3/3	3/3	766
Sodium alginate	200 mg	2	3	−101.5	73.3	0/3	0/3	903
Si (−325 mesh)	200 mg	2	3	−199.7	0.5	0/3	0/3	31
AlSi10Mg	1 g	2	3	+1.9	6.6	3/3	2/3	65
Si (−110/+200 mesh)	1 g	2	3	+0.2	2.8	3/3	3/3	128
Sodium sulfate	1 g	2	3	−4.2	1.1	3/3	3/3	86
Calcium lactate	1 g	1	3	−26.5	9.9	3/3	0/3	263
Xanthan gum	1 g	1	3	−32.9	9.6	3/3	0/3	193
NaCl	1 g	1	6	−1.9	5.3	6/6	5/6	186
CMC	1 g	1	3	−43.2	5.4	3/3	0/3	668
White rice flour	1 g	1	3	−137.9	23.5	0/3	0/3	822
Sodium alginate	1 g	1	3	−291.5	18.2	0/3	0/3	840
Brown rice flour	1 g	1	3	−999.1	0.9	0/3	0/3	7
Si (−325 mesh)	1 g	1,2	6	−999.5	0.7	0/6	0/6	8

S6 Run inventory

Table S5 lists every run recorded in the two test rounds, including runs that were aborted, postponed, or excluded, so that each number in the main text can be traced to the run it came from.

The test battery’s raw records store tilt as a servo setting (0, 45, or 90) rather than as an angle. The tilt plate is geared 2:1, so these settings are tube angles of 0°, 22.5°, and 45° above horizontal, and 45° is the steepest tilt reached in any test. The main text and its figures report tube angles; the conversion is made once, in `paper/figures/make_data_figures.py`.

Table S5: Every recorded run in both test rounds, in date order. *Protocols* lists the test protocols the run requested (Table 3 of the main text; protocol F never produced data); *Doses* is the number of closed-loop doses attempted. Locations: an open laboratory bench (to 6 August 2026), a shared fume hood (11 August to 3 September), and a second fume hood (from 3 September). Runs that failed screening are listed here and in `paper/figures/data/runs_all.csv` with the reason; the Experimental section describes how runs were chosen for each figure.

Start (UTC)	Round	Powder	Protocols	Location	Doses	Screening
2026-08-04 20:43	1	Brown rice flour	ABCDEFGF	open bench	3	excluded: nothing delivered, auger suspected
2026-08-04 21:14	1	White rice flour	ABCDEFGF	open bench	3	passed
2026-08-04 22:49	1	Brown rice flour	ABCDEFGF	open bench	3	excluded: nothing delivered, auger suspected
2026-08-05 14:57	1	Sodium alginate	ABCDEFGF	open bench	3	passed
2026-08-05 18:53	1	Brown rice flour	ABCDEFGF	open bench	3	passed (slow delivery)
2026-08-05 20:00	1	Calcium lactate	ABCDEFGF	open bench	3	passed
2026-08-05 21:12	1	—	—	open bench	0	aborted before testing
2026-08-05 21:52	1	CMC	ABCDEFGF	open bench	3	passed
2026-08-06 14:02	1	Xanthan gum	ABCDEFGF	open bench	3	passed
2026-08-06 14:51	1	NaCl	ABCDEFGF	open bench	3	passed
2026-08-11 18:30	1	AlSi10Mg	ABCDEFGF	fume hood 1	3	excluded: tilt servo failed
2026-08-12 21:51	1	NaCl	ABCDEFGF	fume hood 1	3	passed
2026-08-20 17:56	1	NaCl	ABCDE	fume hood 1	0	excluded: deliberate bench stress test
2026-08-20 19:22	1	Sodium sulfate	ABCDE	fume hood 1	0	passed
2026-08-20 21:48	1	Si (−110/+200 mesh)	ABCDE	fume hood 1	0	passed
2026-08-21 15:13	1	AlSi10Mg	ABCDE	fume hood 1	0	passed
2026-08-21 16:18	1	Si (−325 mesh)	ABCDEG	fume hood 1	3	passed (slow delivery)
2026-08-21 17:07	1	Barium chloride	ABCDE	fume hood 1	0	passed
2026-08-21 18:27	1	Fumed silica	ABCDE	fume hood 1	0	excluded: nothing delivered, outlet not checked
2026-08-21 19:20	1	NaCl	ABCDE	fume hood 1	0	passed
2026-09-03 16:35	2	NaCl	H	fume hood 1	6	excluded: balance could not be read
2026-09-03 17:04	2	NaCl	H	fume hood 1	6	excluded: balance could not be read
2026-09-03 21:40	2	NaCl	H	fume hood 2	6	passed
2026-09-04 00:18	2	NaCl	G	fume hood 2	1	demonstration only
2026-09-04 19:00	2	White rice flour	H	fume hood 2	6	passed
2026-09-04 20:37	2	Xanthan gum	H	fume hood 2	6	passed
2026-09-08 17:27	2	CMC	H	fume hood 2	6	excluded: outlet misaligned
2026-09-08 19:42	2	CMC	H	fume hood 2	6	passed
2026-09-09 18:53	2	Sodium alginate	H	fume hood 2	6	passed
2026-09-09 22:52	2	Calcium lactate	H	fume hood 2	5	passed
2026-09-10 13:39	2	Sodium sulfate	GH	fume hood 2	9	passed
2026-09-10 17:08	2	Barium chloride	GH	fume hood 2	9	excluded: powder caked in the tube
2026-09-10 19:44	2	AlSi10Mg	GH	fume hood 2	9	passed
2026-09-10 21:29	2	Si (−325 mesh)	GH	fume hood 2	9	passed (slow delivery)
2026-09-10 23:19	2	—	—	fume hood 2	0	postponed: failed the stability check
2026-09-15 02:55	2	Si (−110/+200 mesh)	GH	fume hood 2	9	passed
2026-09-15 16:42	2	NaCl	G	fume hood 2	1	demonstration only

S7 AI usage log

In line with *Digital Discovery*’s guidance on the use of LLMs, all AI interactions are preserved publicly in the repository:

- **CAD and code generation:** every coding-agent session (GitHub Copilot coding agent and Claude-family models, 2026) is recorded as a pull request containing the prompt (issue text and review comments), the generated CAD code, rendered views, and the human review thread.
- **Design reviews and literature queries:** Edison Scientific query scripts, task identifiers, raw answers, and extracted

references are archived with the corresponding branches; the consolidated bibliography, with the verification status of each entry, is `paper/references_validation.md`.

- **Text-to-CAD evaluations:** CADSmith run artifacts and the Zoo Design Studio (Zookeeper agent) session transcripts and notes are preserved on their branches.
- **Analysis and writing:** the data-analysis scripts behind every figure, and the AI-drafted revisions of this manuscript, are in `paper/` with their review threads.
- **Qualitative observations:** the team's running journal of AI behaviour, good and bad, is discussion #39 in the repository.

Fig. S6 shows how agent use was spread over the project. Team members addressed the two coding agents 754 times between April and September 2026, and the agents authored 789 of the 877 commits (90%). The counts cover all work in the repository (CAD, firmware, data analysis, and writing), not only the design of the parts. The team moved from the GitHub Copilot coding agent to Claude at the start of July 2026.

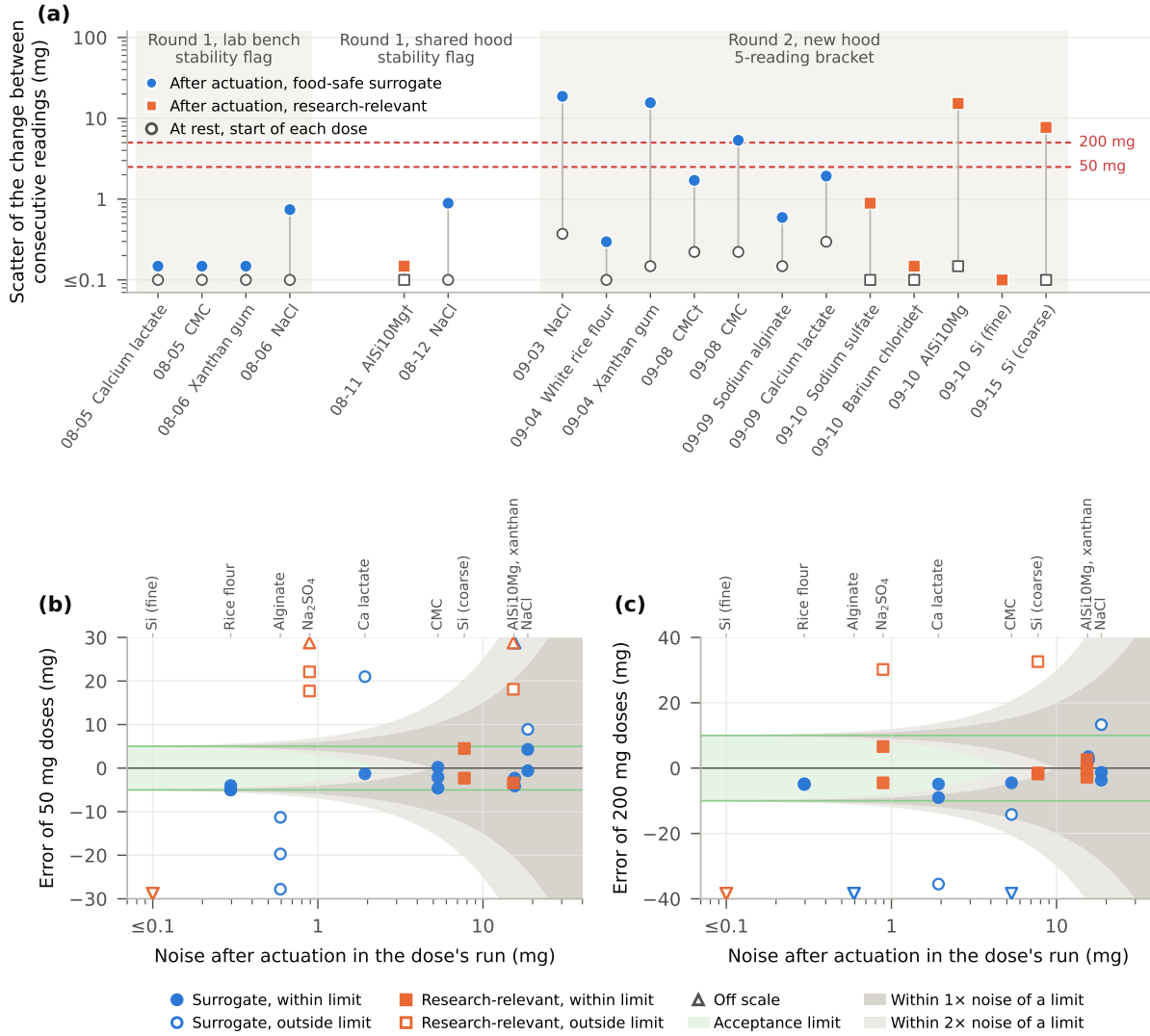


Figure S5: Balance noise after actuation and the pass/fail calls of small doses. (a) Filled markers: noise after actuation for every run with a closed-loop tap phase, the robust standard deviation of the change between consecutive tap-phase readings, each taken 1.5 s after two solenoid taps with the tube horizontal. Open markers apply the same statistic at rest, to the first reading of each dose minus its tare reading. Fine silicon and barium chloride never reached the tap phase, so their values come from fine-phase steps that delivered nothing. Values below 0.1 mg are drawn at 0.1 mg, and † marks runs excluded from the dose analysis. Above the dashed lines, no 50 mg or 200 mg dose can be a robust pass. (b, c) Error of each valid 50 and 200 mg dose against the noise of its run (named along the top). Light and dark grey zones lie within two and one times the noise of a limit. Markers as in Fig. 5 of the main text.

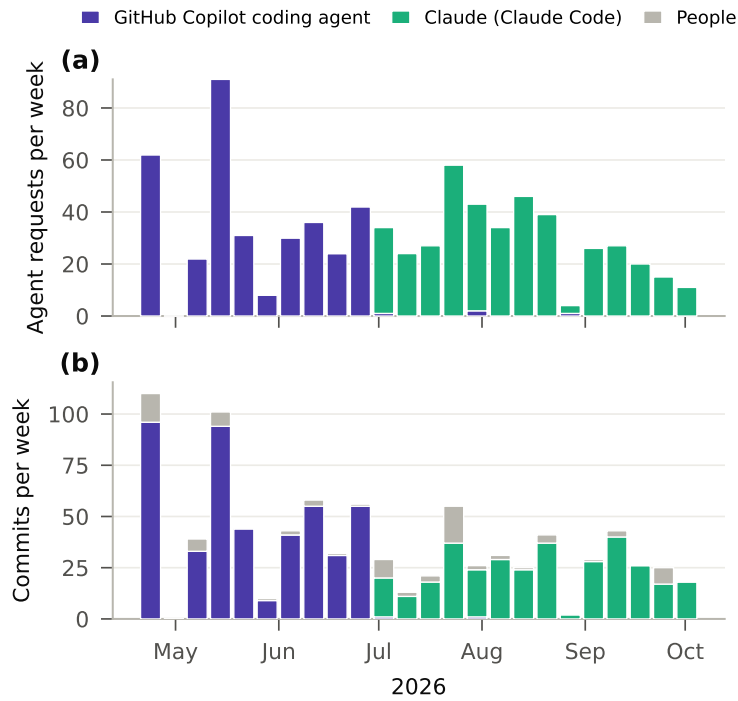


Figure S6: Generative-AI agent activity per week over the project. (a) Requests to an agent: comments and issue or pull-request descriptions written by a team member that address @copilot or @claude, plus pull requests the Copilot coding agent opened when assigned an issue. Most such requests started one agent session, so the count is an approximate measure of use. (b) Commits on all branches by author; commits that were later rebased onto another branch are counted once. Counts are from the public repository history ([paper/figures/data/build_ai_usage.py](https://github.com/paper/figures/data/build_ai_usage.py)).